Height Growth Equations for CONUS: Red Maple

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2026-03-03

Data

We took the FIA height growth data subset for red maple, filtered out FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 217592 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Northern Lakes and Forests	5.2.1	48286
Northern Appalachian and Atlantic Maritime Highlands	5.3.1	25962
Acadian Plains and Hills	8.1.8	11265
Southeastern Plains	8.3.5	10750
Ridge and Valley	8.4.1	10374
Central Appalachians	8.4.2	10240
North Central Hardwood Forests	8.1.4	10168
Piedmont	8.3.4	9828
North Central Appalachians	5.3.3	9761
Blue Ridge	8.4.4	9744
Western Allegheny Plateau	8.4.3	8480
Northern Allegheny Plateau	8.1.3	8415
Northeastern Coastal Zone	8.1.7	7579
Middle Atlantic Coastal Plain	8.5.1	5908
Southwestern Appalachians	8.4.9	4057
Erie Drift Plain	8.1.10	3854
Eastern Great Lakes Lowlands	8.1.1	3721
Interior Plateau	8.3.3	2689
Southern Michigan/Northern Indiana Drift Plains	8.1.6	2510
South Central Plains	8.3.7	2188

NA_L3NAME	NA_L3CODE	n
Huron/Erie Lake Plains	8.2.2	1616
Driftless Area	8.1.5	1551
Southern Coastal Plain	8.5.3	1447
Northern Piedmont	8.3.1	1140
Interior River Valleys and Hills	8.3.2	1028
Eastern Corn Belt Plains	8.2.4	717
Ouachita Mountains	8.4.8	695
Atlantic Coastal Pine Barrens	8.5.4	597
Ozark Highlands	8.4.5	570
Boston Mountains	8.4.6	565
Southeastern Wisconsin Till Plains	8.2.1	498
Mississippi Valley Loess Plains	8.3.6	428
Northern Minnesota Wetlands	5.2.2	315
Arkansas Valley	8.4.7	245
Mississippi Alluvial Plain	8.5.2	228
Western Corn Belt Plains	9.2.3	50
NA	NA	41
Central Corn Belt Plains	8.2.3	37
Western Gulf Coastal Plain	9.5.1	22
East Central Texas Plains	8.3.8	10
Lake Manitoba and Lake Agassiz Plain	9.2.2	7
Central Irregular Plains	9.2.4	5
Southern Florida Coastal Plain	15.4.1	1

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 217592 growth observations, estimating β_0 through β_6. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for red maple. In our data that value is: 135 feet.

The parameter estimates and fit statistics are:

	Coef.	Std. error	t-stat.	p
b1	0.0170741	0.0002654	64.3340327	0.0000000
b2	1.8614481	0.0245321	75.8779359	0.0000000
b3	0.0221716	0.0066010	3.3588072	0.0007829
b4	0.0011908	0.0000320	37.1742238	0.0000000
b5	-0.0000301	0.0000385	-0.7813099	0.4346211
b6	0.0005797	0.0000337	17.1975773	0.0000000
b7	0.0022705	0.0002369	9.5823798	0.0000000

```
Residual Standard Error: 5.08204561223135 on 217585 degrees of freedom. AIC =  1324993
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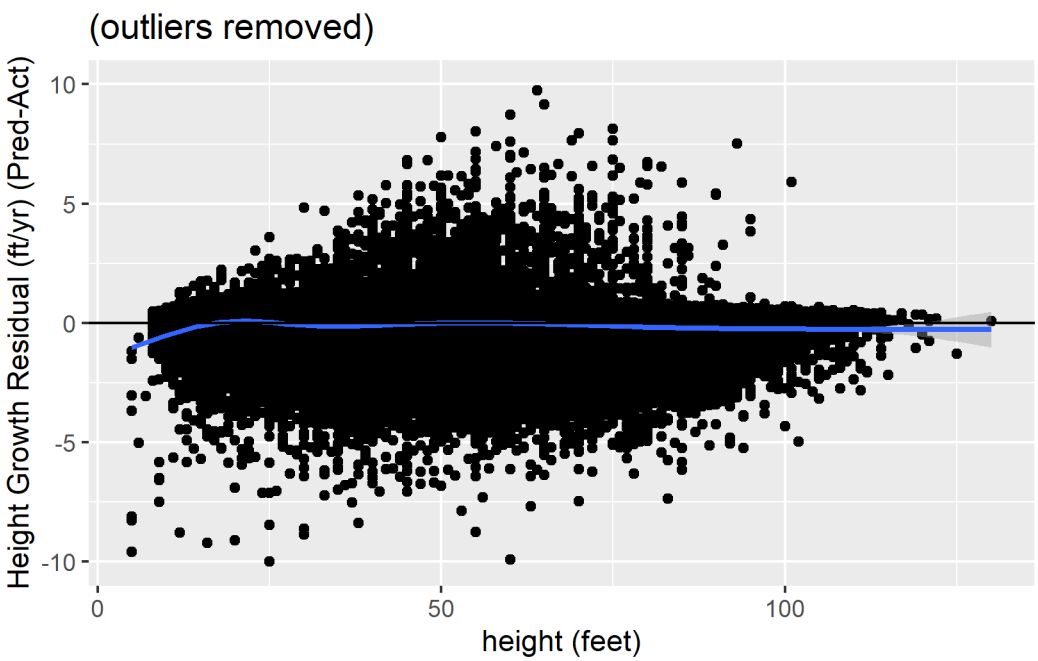
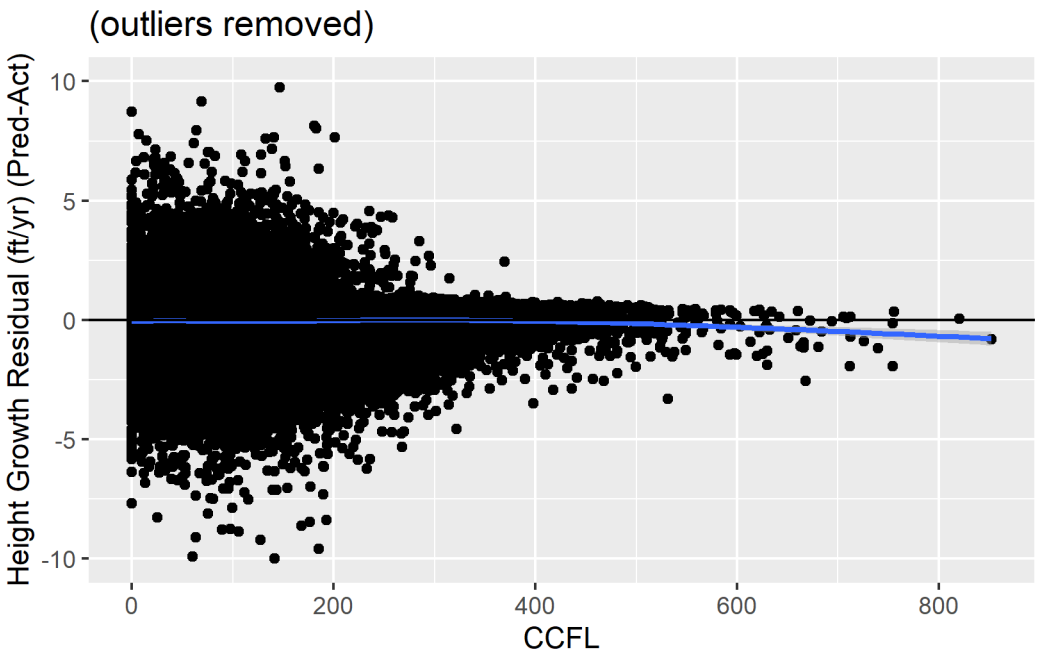
Residual Analysis for Equation 1

Mensurational Variables

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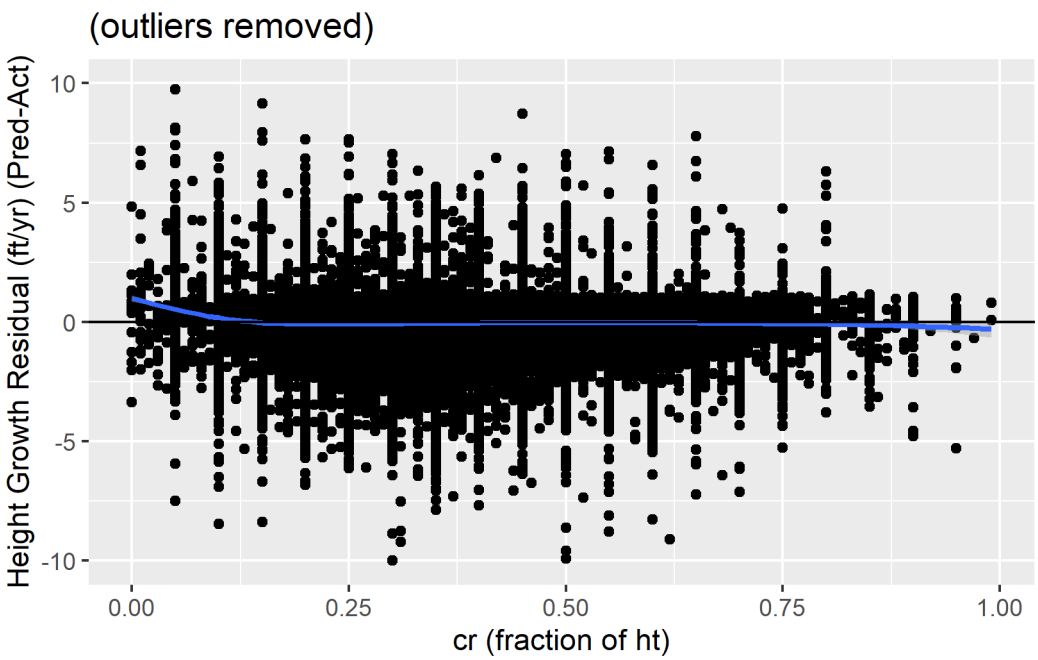
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
200
400
600
800
CCFL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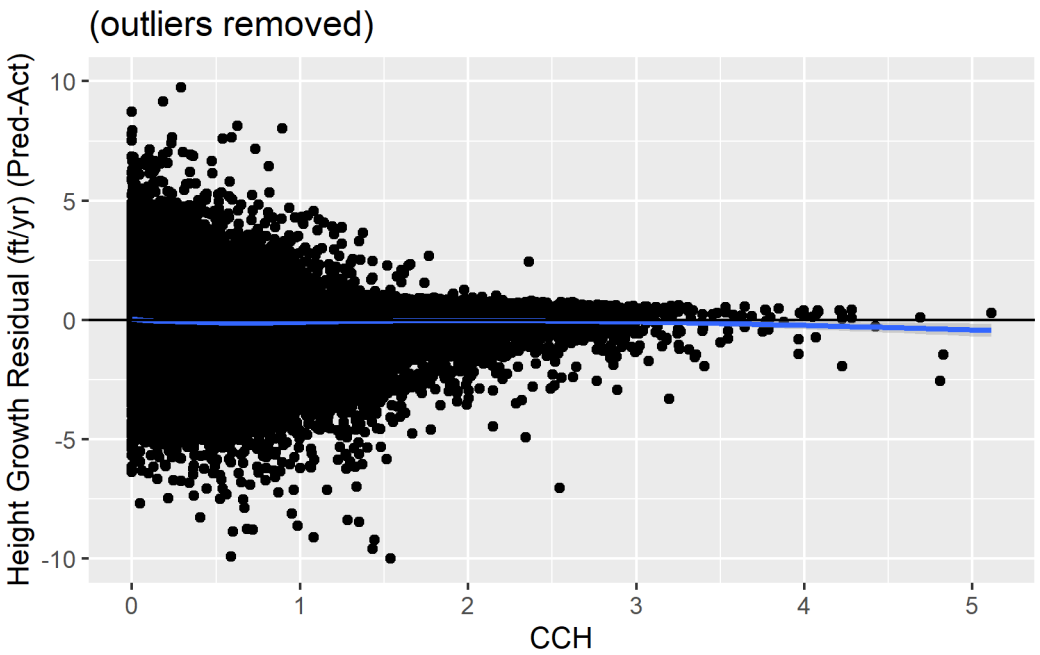
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
cr (fraction of ht)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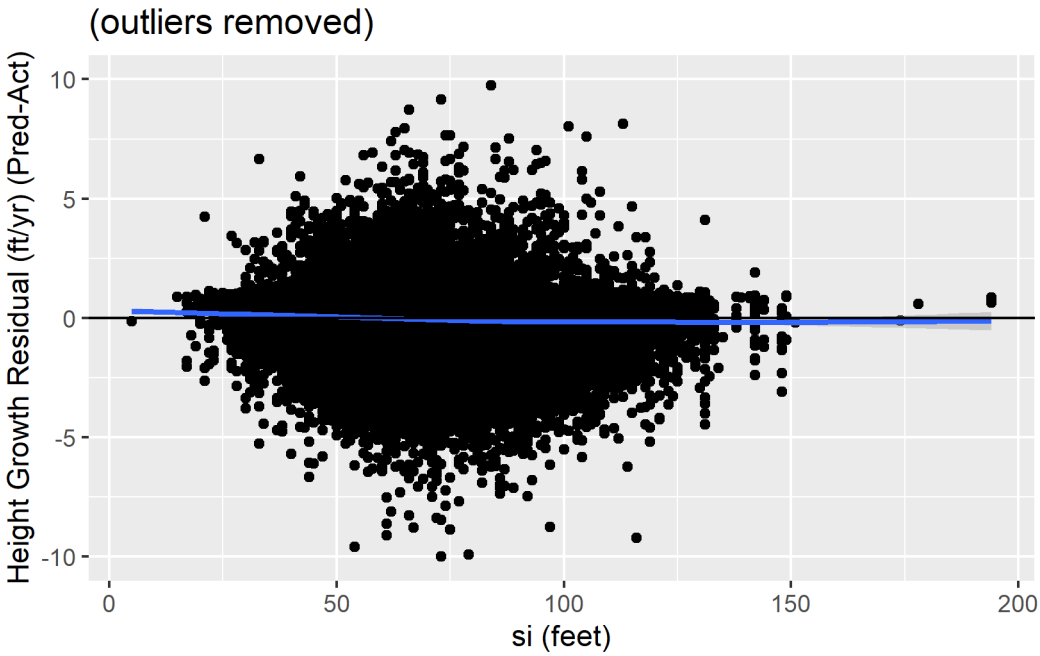
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
1
2
3
4
5
CCH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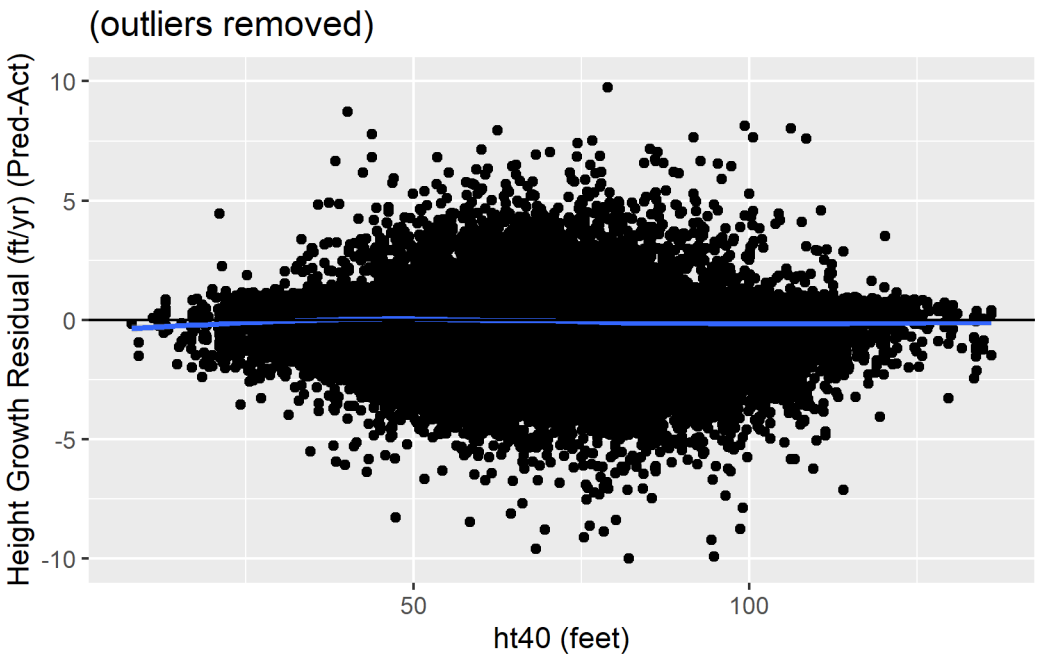
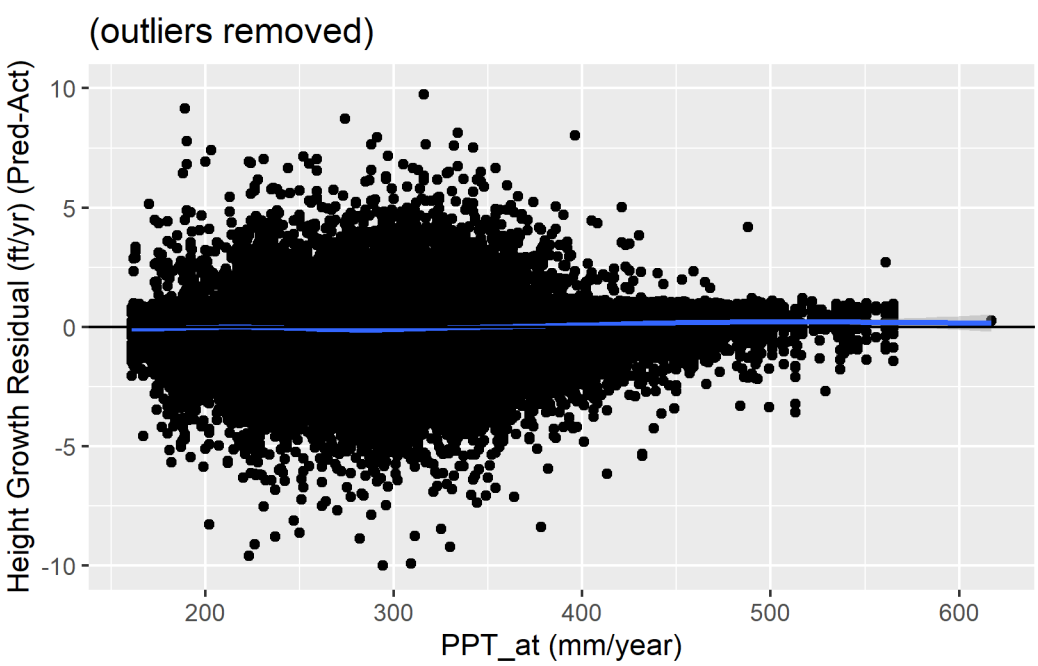
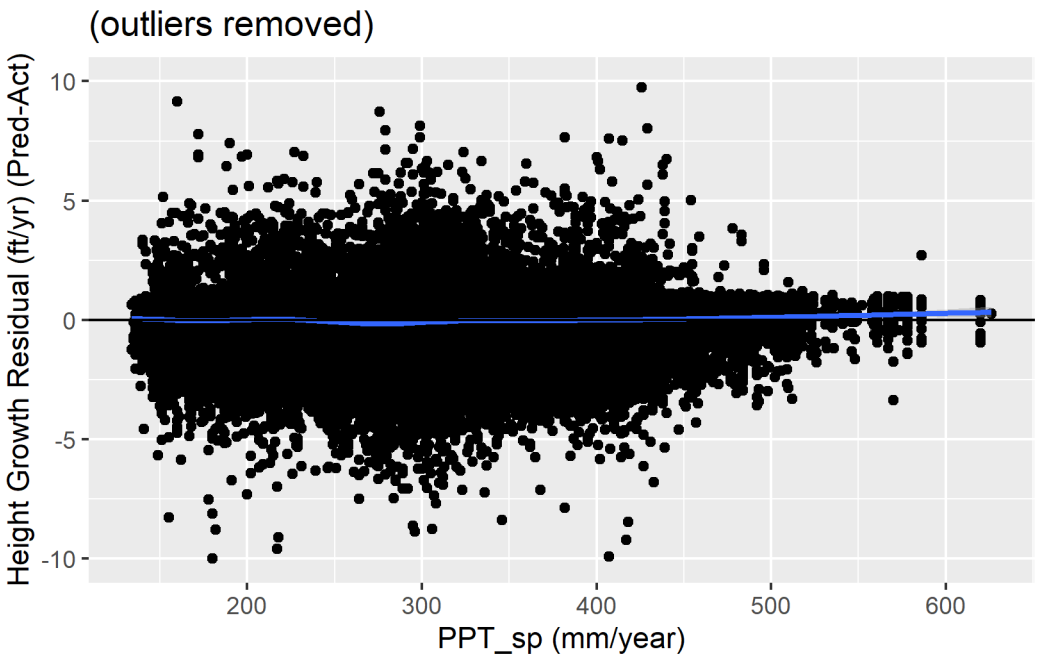
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
50
100
150
200
si (feet)



Environmental and Climate Variables

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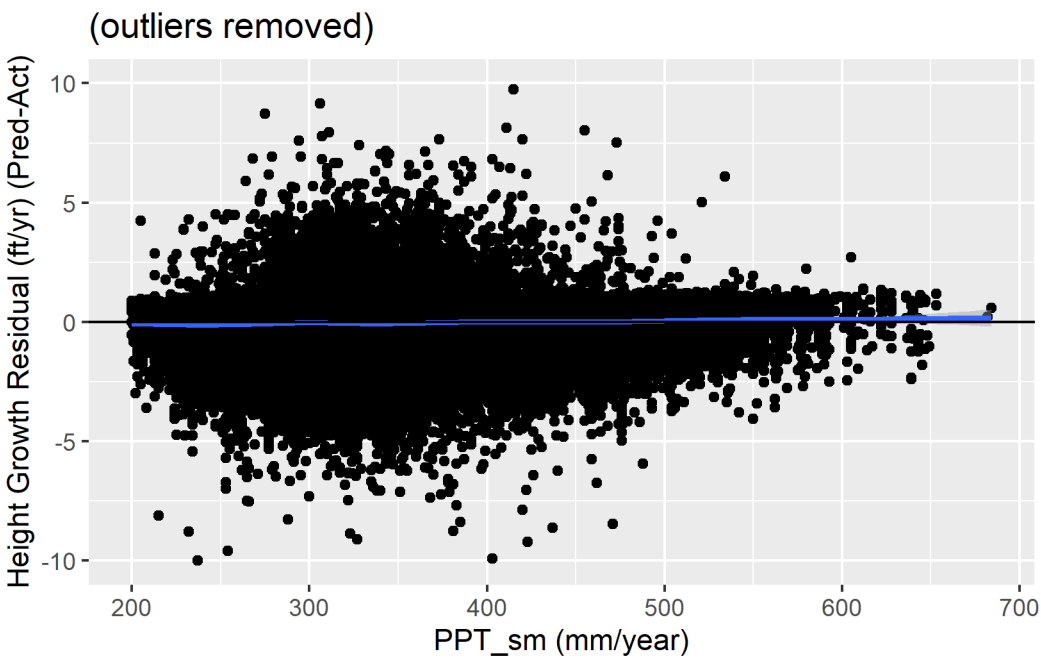
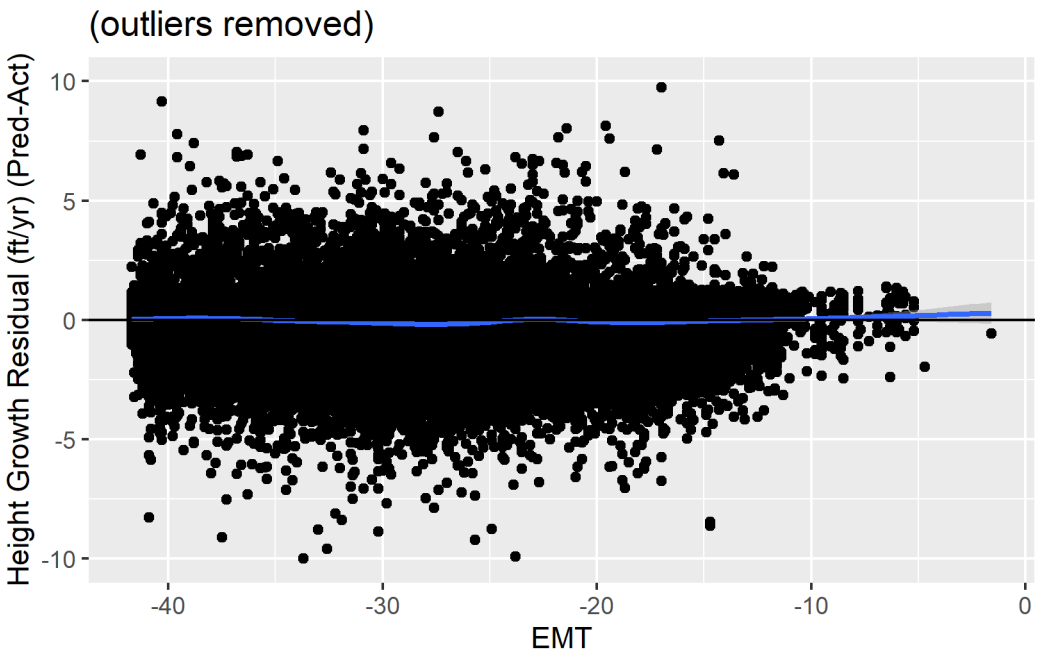
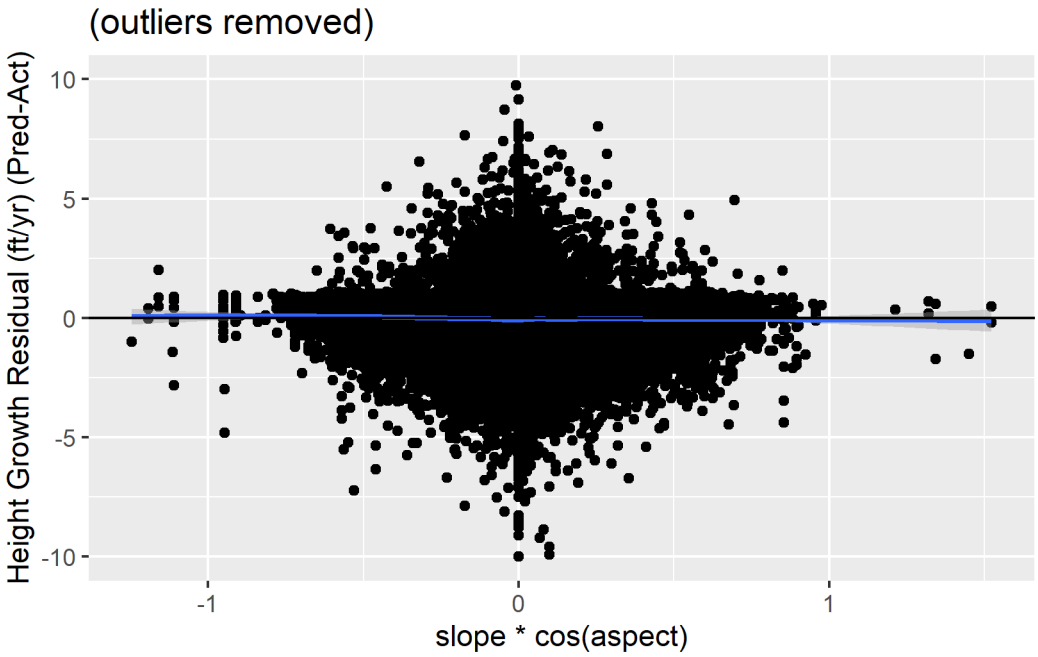
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
200
300
400
500
600
PPT_sp (mm/year)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
200
300
400
500
600
700
PPT_sm (mm/year)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-40
-30
-20
-10
0
EMT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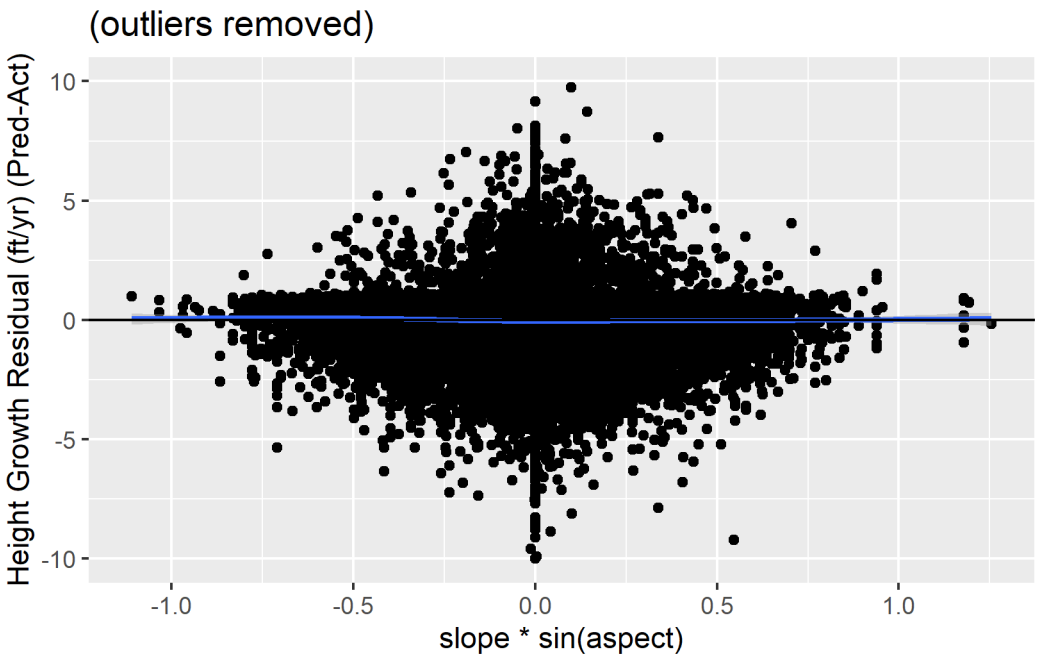
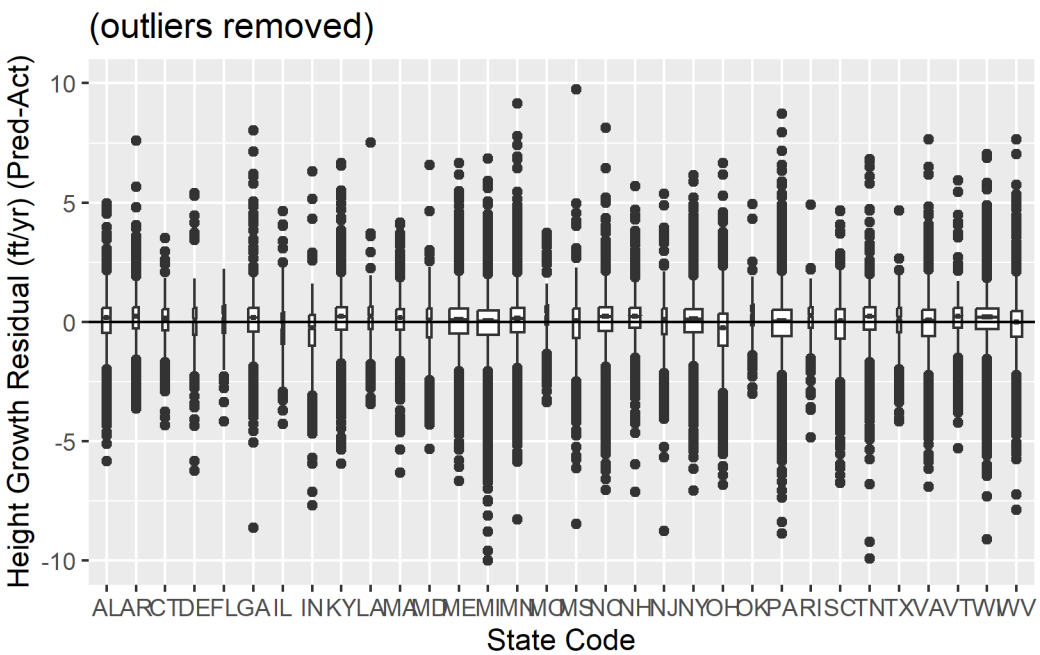
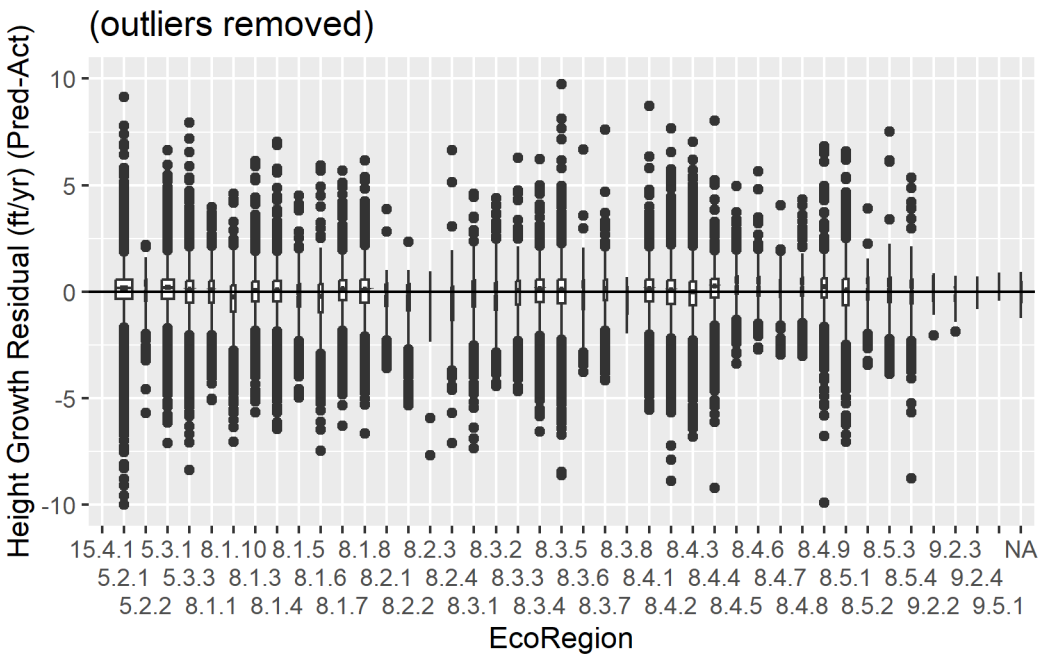
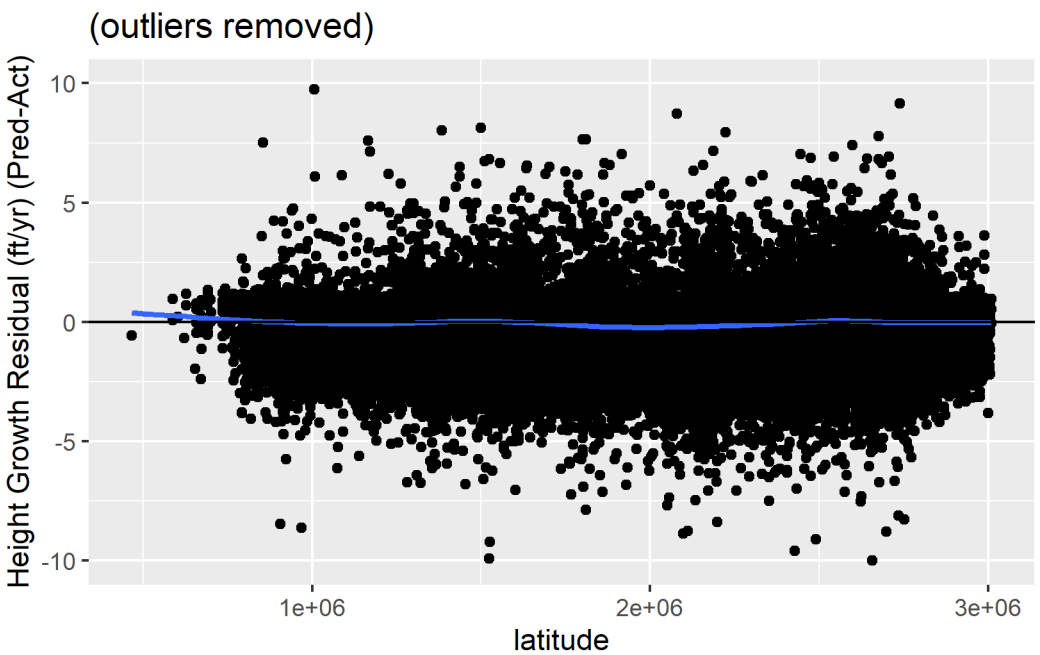
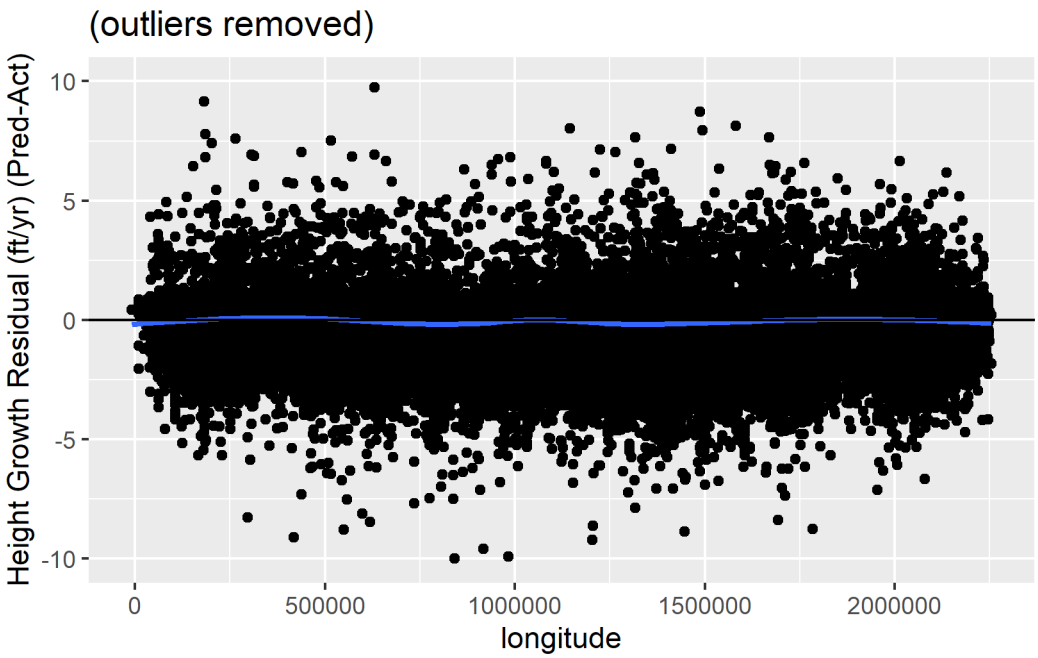
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1
0
1
slope * cos(aspect)

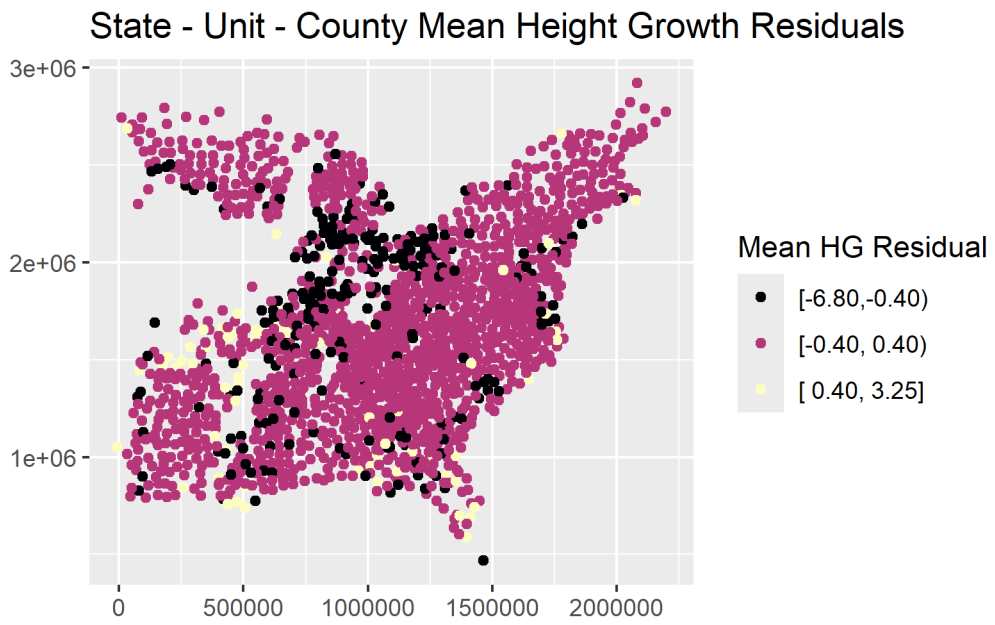

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
AL AR CT DE FL GA IL IN KY LA MA MD ME MI MN MO MS NC NH NJ NY OH OK PA RI SC TN TX VA VT WI WV
State Code

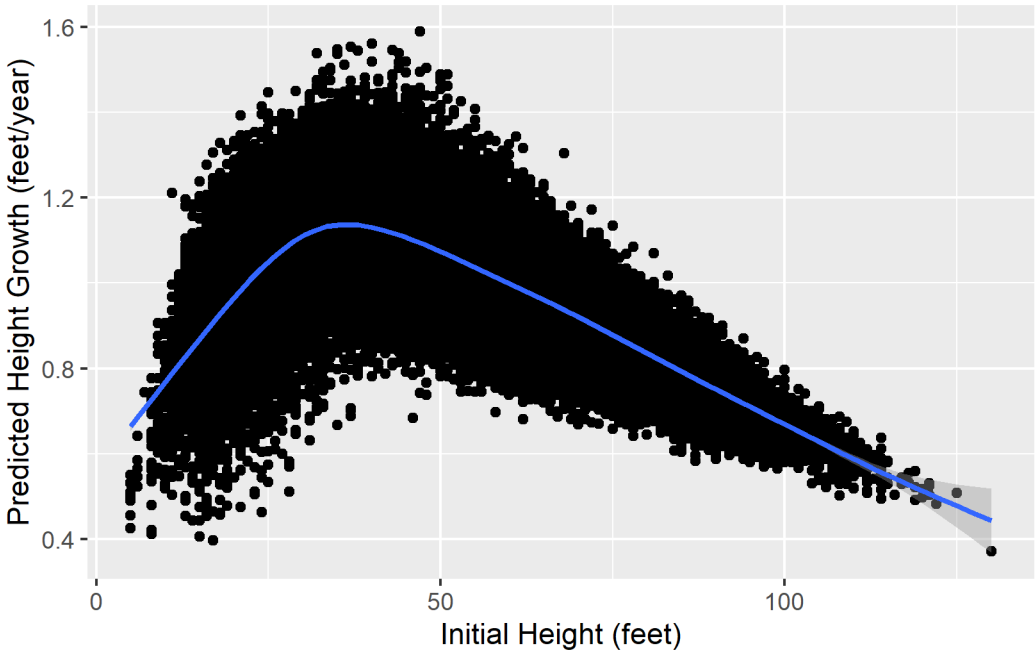

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
15.4.1 5.3.1 8.1.10 8.1.5 8.1.8 8.2.3 8.3.2 8.3.5 8.3.8 8.4.3 8.4.6 8.4.9 8.5.3 9.2.3 NA
5.2.1 5.3.3 8.1.3 8.1.6 8.2.1 8.2.4 8.3.3 8.3.6 8.4.1 8.4.4 8.4.7 8.5.1 8.5.4 9.2.4
5.2.2 8.1.1 8.1.4 8.1.7 8.2.2 8.3.1 8.3.4 8.3.7 8.4.2 8.4.5 8.4.8 8.5.2 9.2.2 9.5.1
EcoRegion

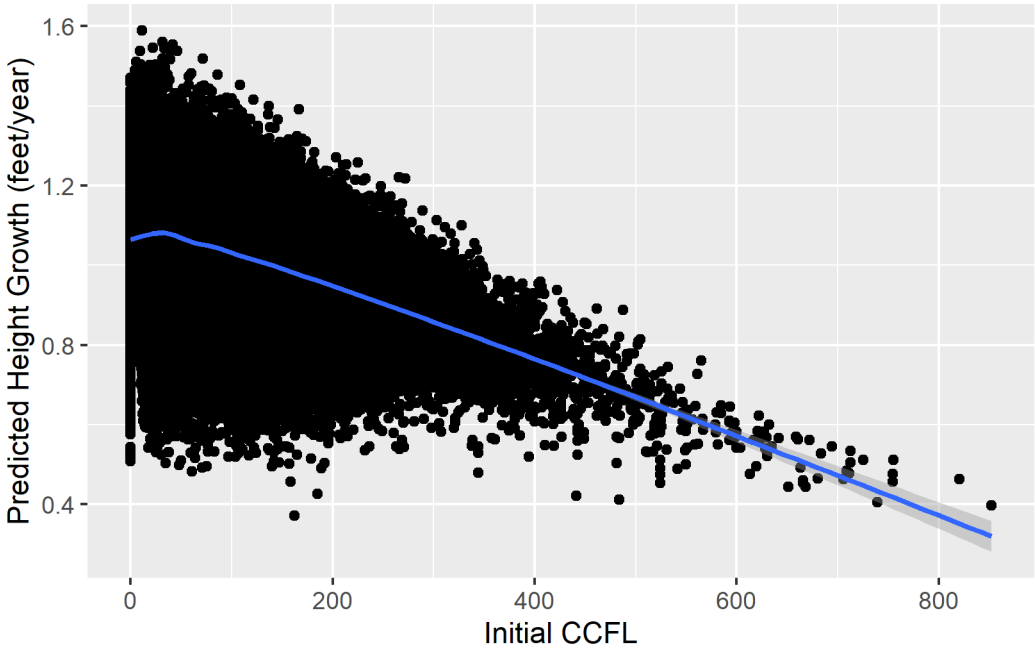

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
1e+06
2e+06
3e+06
latitude

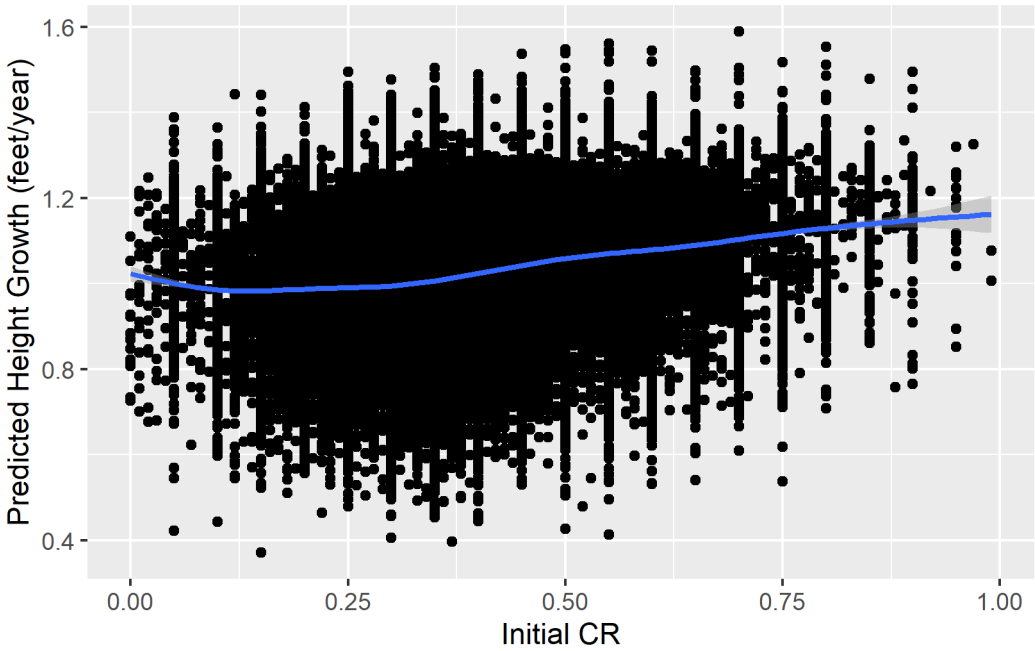

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
10
5
0
-5
-10
0
500000
1000000
1500000
2000000
longitude

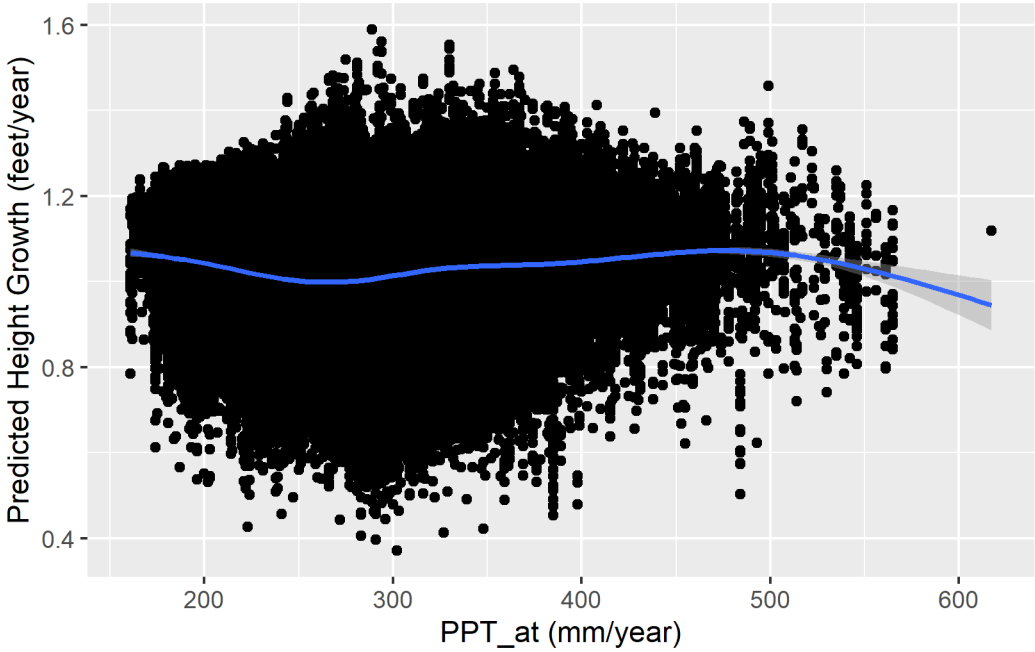

State - Unit - County Mean Height Growth Residuals
3e+06
2e+06
1e+06
0
500000
1000000
1500000
2000000
Mean HG Residual
[-6.80,-0.40)
[-0.40, 0.40)
[0.40, 3.25]

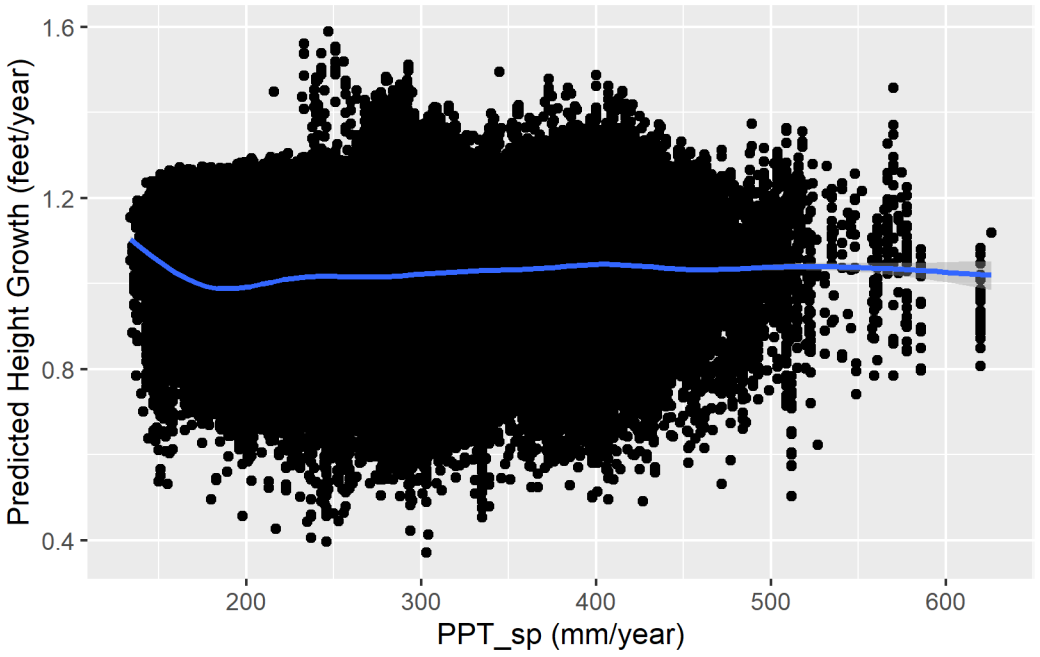
Prediction Performance

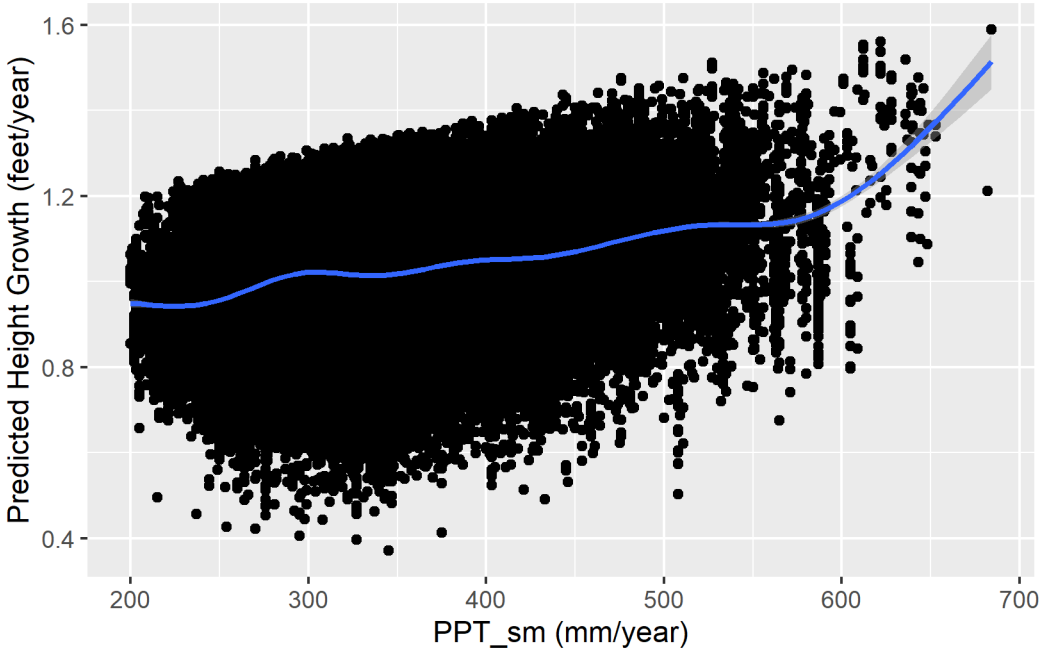


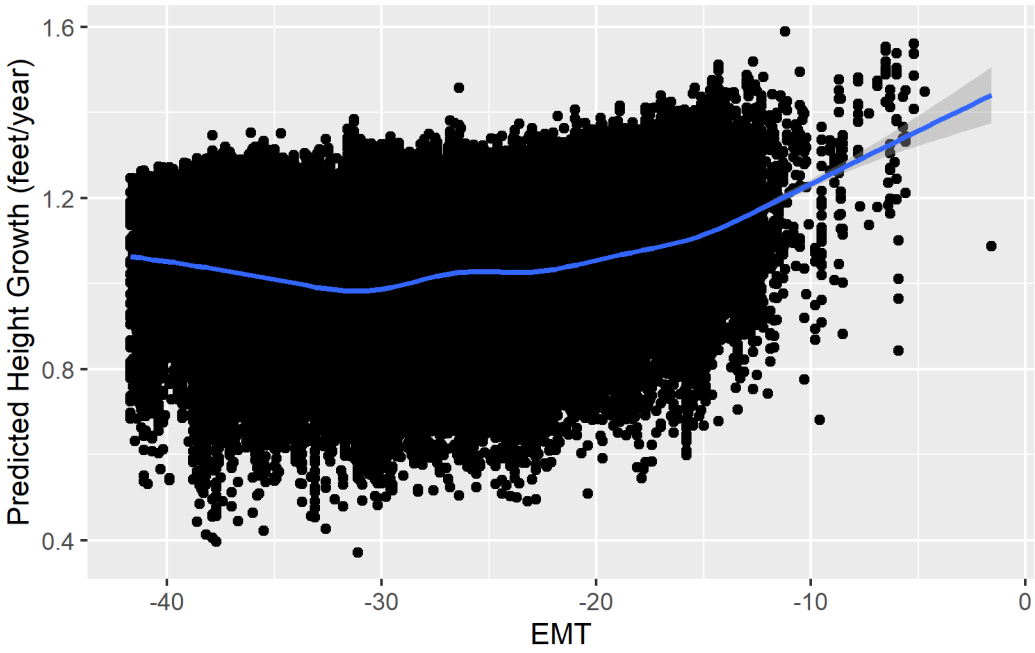



Predicted Height Growth (feet/year)
1.6
1.2
0.8
0.4
0.00
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
1.6
1.2
0.8
0.4
200
300
400
500
600
PPT_at (mm/year)


Predicted Height Growth (feet/year)
1.6
1.2
0.8
0.4
200
300
400
500
600
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
1.6
1.2
0.8
0.4
200
300
400
500
600
700
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
1.6
1.2
0.8
0.4
-40
-30
-20
-10
0
EMT